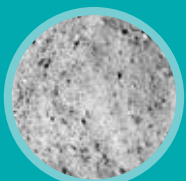


SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

Right from exploring the Fairy Circles in Namibia, to tackling the question of whether VR can actually reduce pain, we have it all this month in science. We also explore how the dinosaurs actually met their end.



OSIRIS-REx discovers water on Bennu

The spacecraft on a sample return mission to the asteroid has identified signs of hydroxyls - atoms of oxygen and hydrogen that are bonded together - in water bearing clay minerals all over the asteroid. The hydrated minerals are actually a part of the asteroid, and have not been deposited there by an impactor.

<https://dgit.in/h2obennu>

WHAT'S NEW

Worm brain taught to park car

Caenorhabditis elegans is one of the simplest microorganisms around. Within its tiny brains, are 300 cells which have about 7,000 neurons between them. The wiring diagram of its brain, or the connectome, is available as a virtual copy in its entirety. Researchers from TU Wien (Vienna) and MIT trained a simple neural network to park a car (actually a robotic rover). When compared to other AI systems that performed the same task like finding a parking space, avoiding running into obstacles, and navigating the car to the parking spot, the C. elegans-based system was superior in many aspects. It was also more resilient to external noise, and could perform better even



when unexpected disturbances were deliberately introduced. The most important advantage though, is that the researchers could easily understand what was going on in the system, and what the AI was doing at any point of time.

<https://dgit.in/wbcar>

LIGO detects biggest black hole collision

The Advanced Laser Interferometer Gravitational-wave Observatory (LIGO) is a facility dedicated to detect ripples in the spacetime continuum, known as gravitational waves. The LIGO facility has been used for two observation campaigns so far. The first one ran from 12 September 2015 to 19 January 2016. During this time, three black hole collisions were detected. The second observation run lasted from 30



November 2016 to 25 August 2017. During this run, the collision of two neutron stars was detected. Three additional black hole mergers were detected during the second run. Now, scientists combing through the

data from the second run using better detection techniques, have announced the detection of four additional collision events, during the second run. One of these is the biggest known black hole merger, in a collision that occurred nine billion light years away. The gravitational waves from the event was detected on 29 July, 2017, and is officially designated as GW170729.

<https://dgit.in/bigligo>



CRISPR babies

There are unsubstantiated claims that Chinese researchers have edited baby genes using CRISPR. The exercise has raised ethical concerns. There were problems with acquiring consent, they did not execute the editing well, and there is no way to say if it was a successful operation.

<https://dgit.in/crisprbabies>



Smallest 3D transistor

Researchers from MIT have produced the smallest transistor yet, at half the price of commercial models. The new method modifies a semiconducting substrate on an atom by atom basis. The technology can be used to make smaller and more efficient electronics.

<https://dgit.in/microtr>



The Great Dying

252 million years ago, at the end of the Permian period, was the most severe extinction event in Earth's history, killing off 96% of marine and 70% of terrestrial life. New research indicates the extinction was caused by a period of global warming due to a series of massive volcanic eruptions in Siberia.

<https://dgit.in/grtdyng>